

M.Sc. in Electronics and Advanced Materials for Renewable Energies



BOUDHAIR HAMZA

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in Hamza Boudhair

Profile

Motivated M.Sc. graduate in Electronics and Advanced Materials for Renewable Energies, with strong expertise in electrical engineering (LV/MV/HV), energy efficiency, aerodynamic simulation, and renewable energy system analysis. Through academic and professional experience in research environments (ENSA Agadir, ECOWATT, Regional Energy Directorate), I developed solid skills in electrical system design, techno-economic evaluation, wind turbine aerodynamic modelling (QBlade), and optimization of PV and hybrid systems.

Highly rigorous, autonomous, and research-driven, I am seeking to contribute to advanced projects in renewable energy engineering, computational modelling, and sustainable energy research within international laboratories or academic institutions.

Education

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| 2023 – 2025 | Master of Science, Polydisciplinary Faculty of Ouarzazate Electronics and Advanced Materials for Renewable Energies.
Main focus: electrical engineering, renewable energy systems, and advanced energy materials. |
| 2021 – 2022 | Professional Bachelor's Degree, Higher School of Technology, Essaouira Renewable Energies and Sustainable Development.
Focus: renewable energy systems, energy audits, and PV installation. |
| 2019 – 2021 | University Diploma of Technology, Higher School of Technology, Guelmim Renewable Energy and Energy Efficiency.
Emphasis: PV and wind systems, electrotechnics, and energy efficiency. |
| 2018 | Baccalaureate in Physical Sciences. |

Professional Experience

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| Feb. – Jun. 2025 | National School of Applied Sciences (ENSA), Agadir – Energy, Mechanics and Environment Laboratory.
<i>Research Topic:</i> Numerical investigation of a horizontal-axis wind turbine using QBlade. Developed aerodynamic modelling based on Blade Element Momentum Theory (BEMT), performed multi-parameter optimisation of blade geometry, and analysed performance curves under varying operating conditions. The work involved computational simulation, aero-mechanical interpretation, and validation of numerical indicators relevant to wind energy research.
<i>Report:</i> Consult the report |
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Professional Experience (continued)

- Apr. – Jun. 2022 **ECOWATT Company, Agadir.**
Research Topic: Techno-economic assessment and modelling of a grid-connected photovoltaic system. Conducted simulation campaigns using PVSYST, evaluated energy production profiles, modelled losses, assessed investment indicators (NPV, IRR), and produced research-level technical documentation. Work contributed to performance optimisation of PV systems under Moroccan climatic conditions.
Report: Consult the report
- Apr. – Jun. 2021 **Regional Directorate of Energy and Mines, Agadir.**
Research Topic: Modelling and sizing of standalone and hybrid photovoltaic systems. Developed load profiles, designed hybrid PV architectures, analysed operational scenarios, and proposed optimised technical configurations for rural and semi-isolated zones. Integrated energy demand analysis and system performance forecasting.
- Mar. – Apr. 2021 **Higher School of Technology, Guelmim.**
Research Topic: Simulation and optimisation of a photovoltaic pumping system. Conducted feasibility assessment, performed numerical modelling under variable operating conditions, and evaluated system reliability and performance metrics for off-grid hydraulic applications.

Technical Skills

Electrical and Power Engineering: Advanced design and modelling of electrical systems (LV/MV/HV); grid integration principles; high-voltage engineering; protection systems; normative compliance (NF C15-100); development of detailed electrical documentation.

Energy Systems Engineering: Modelling, simulation, and optimization of renewable energy systems; energy efficiency evaluation; techno-economic assessment; hybrid system integration; performance and sensitivity analyses.

Power Electronics and Control: Design and operation of converters, electric drives, and power systems; implementation of control algorithms and energy conversion strategies.

Project Coordination and Scientific Reporting: Management of engineering workflows, coordination of interdisciplinary tasks, preparation of research deliverables, and production of technical/scientific reports.

Communication and Scientific Dissemination: Technical and scientific presentations, collaborative research, interdisciplinary communication, and client-facing technical consultation.

Software Platforms: AutoCAD, Revit (BIM), Caneco, Dialux, PVSYST, PV*SOL, QBlade, RETScreen, TRNSYS, MATLAB, Python, Microsoft Office.

Certifications

- 2023 **Formation Logiciel PVSYST et PV*SOL** — Techniques de dimensionnement et simulation photovoltaïque.
- 2022 **Certification en MATLAB** — Programmation appliquée aux systèmes énergétiques.

Languages

Arabic: Native proficiency | **English:** Full professional proficiency | **French:** Full professional proficiency

